

Taxonomy Reconciliation QA Review

GBIF Gap Analysis — Verifying 09a Matching Results

gbifgaps

February 13, 2026

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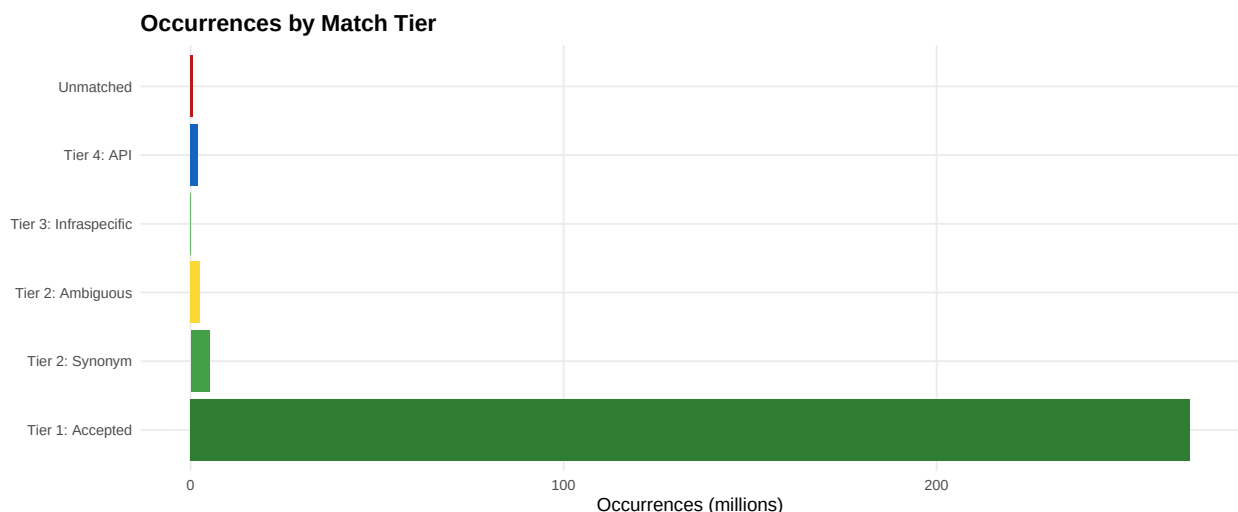
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Overview

This report reviews the taxonomy reconciliation produced by `09a_reconcile_taxonomy.R`. It helps you verify match quality and identify species that could be resolved with manual overrides.

Reconciliation summary: 55,357 of 67,668 GBIF species matched to the national backbone (81.8% of species, 99.8% of occurrences).

Tier	Type	Species	Occurrences	% Species	% Occurrences
tier1	accepted_name	49,931	267,850,375	73.8	96.33
tier2	synonym	3,473	5,041,544	5.1	1.81
tier2_ambiguous	synonym	236	2,595,089	0.3	0.93
tier3	infraspecific_collapse	38	1,582	0.1	0.00
tier4	gbif_api_synonym	1,679	1,900,586	2.5	0.68
unmatched	below_api_threshold	7,811	27,520	11.5	0.01
unmatched	binomial_no_match	3,799	593,270	5.6	0.21
unmatched	genus_only	693	57,902	1.0	0.02
unmatched	hybrid	6	548	0.0	0.00
unmatched	other	2	184	0.0	0.00



1. Ambiguous Synonym Matches

These species matched a Dyntaxa synonym name that points to **more than one accepted taxon**. The script took the first match. Review these to confirm correctness — especially the high-occurrence ones.

All Ambiguous Matches

Table 2: 236 ambiguous synonym matches

gbif_species	total_ochosen_match	all_candidates	n_candidates
Corvus cornix	2,245,884	Corvus corone cornix	1
Ochlodes venata	83,564	Ochlodes sylvanus	1
Rhodofomes roseus	71,536	Fomitopsis rosea	1

gbif_species	total_	chosen_match	all_candidates	n_candidates
Galium album	34,112	Galium mollugo subsp. erectum	Galium mollugo subsp. erectum	1
Inonotus leporinus	23,914	Pelloporus leporinus	Pelloporus leporinus	1
Plicaturopsis crispa	17,588	Plicatura crispa	Plicatura crispa	1
Xylodon radula	9,612	Basidioradulum radula	Basidioradulum radula	1
Plagiochila asplenioides	8,956	Plagiochila asplenioides s.lat.	Plagiochila asplenioides s.str. Plagiochila asplenioides s.lat.	2
Stellaria apetala	8,274	Stellaria media	Stellaria media Stellaria pallida	2
Radula complanata	7,198	Radula complanata s.lat.	Radula complanata s.str. Radula complanata s.lat.	2
Lenzites betulinus	6,810	Trametes betulina	Trametes betulina	1
Senecio vernalis	6,644	Senecio leucanthemifolius subsp. vernalis	Senecio leucanthemifolius subsp. vernalis	1
Salix fragilis	6,424	Salix × fragilis	Salix × fragilis Salix euxina	2
Bubulcus ibis	5,296	Ardea ibis/coromanda	Ardea ibis/coromanda Ardea ibis	2
Pachykytospora tuberculosa	4,844	Haploporus tuberculosus	Haploporus tuberculosus	1
Mycobilimbia sphaeroides	3,608	Mycobilimbia pilularis	Mycobilimbia pilularis Mycobilimbia carneoalbida	2
Exidia cartilaginea	2,202	Exidia candida var. cartilaginea	Exidia candida var. cartilaginea	1
Apterygida albipennis	1,882	Apterygida media	Apterygida media	1
Sphaerodorum gracilis	1,882	Sphaerodorum flavum	Sphaerodorum flavum	1
Atheta arctica	1,776	Philhygra arctica	Philhygra arctica	1
Scrippsiella hangoei	1,732	Apocalathium malmogiense	Apocalathium malmogiense	1
Atheta hygrobia	1,644	Philhygra hygrobia	Philhygra hygrobia	1
Gymnodinium simplex	1,620	Protodinium simplex	Protodinium simplex	1
Hapalopilus croceus	1,456	Aurantiporus croceus	Aurantiporus croceus	1
Lagis koreni	1,306	Pectinaria koreni	Pectinaria koreni	1
Xestia speciosa	1,230	Xestia baltica	Xestia baltica Xestia arctica	2
Atheta debilis	1,227	Philhygra debilis	Philhygra debilis	1
Mycenastrum corium	1,155	Mycenastrum corium (Lam. & DC.) Desv.	Mycenastrum corium (Lam. & DC.) Desv.	1
Atheta elongatula	1,068	Philhygra elongatula	Philhygra elongatula	1

gbif_species	total_ochosen_match	all_candidates	n_candidates
Hymenoscyphus calyculus	946	Hymenoscyphus virgultorum Hymenoscyphus serotinus	2
Atheta malleus	918	Philhygra malleus	1
Neobulgaria pura	886	Ombrophila pura	1
Bryum pallens	854	Ptychostomum pallens	1
Atheta palustris	843	Philhygra palustris	1
Pachyphiale fagicola	824	Gyalecta fagicola	1
Rhizopogon roseolus	800	Rhizopogon roseolus group	1
Onnia triquetra	766	Pelloporus triqueter	1
Melampsora caprearum	756	Melampsora farinosa	1
Erysiphe cichoracearum	750	Golovinomyces cichoracearum s.lat. Golovinomyces cichoracearum	2
Delphinium ajacis	732	Consolida ajacis Consolida orientalis Consolida ajacis	2
Epichloe typhina	730	Epichloë typhina s. lat. Epichloë typhina	2
Atheta britteni	714	Philhygra britteni	1
Dendrodrilus rubidus	678	Bimastos rubidus	1
Amsinckia menziesii	632	Amsinckia intermedia Amsinckia lycopsoides Amsinckia intermedia Amsinckia retrorsa Amsinckia micrantha	4
Conocephalum conicum	632	Conocephalum conicum s.str. Conocephalum conicum s.lat.	2
Juncus alpinus	628	Juncus alpinoarticulatus subsp. rariflorus Juncus alpinoarticulatus subsp. alpinoarticulatus	2
Antrodia albida	584	Antrodia serpens Antrodia heteromorpha	2
Gowardia nigricans	574	Alectoria nigricans	1
Synedra ulna	572	Ulnaria ulna s.lat. Ulnaria ulna	2
Carex leersii	534	Carex echinata Carex divulsa subsp. leersii Carex echinata	2

Flagged for Review

These ambiguous matches have > **5,000 occurrences** and deserve manual verification:

gbif_species	total_ochosen_match	all_candidates	n_candidates
Corvus cornix	2,245,884	Corvus corone cornix	1

gbif_species	total_oc	chosen_match	all_candidates	n_candidates
Ochlodes venata	83,564	Ochlodes sylvanus	Ochlodes sylvanus	1
Rhodofomes roseus	71,536	Fomitopsis rosea	Fomitopsis rosea	1
Galium album	34,112	Galium mollugo subsp. erectum	Galium mollugo subsp. erectum	1
Inonotus leporinus	23,914	Pelloporus leporinus	Pelloporus leporinus	1
Plicaturopsis crispa	17,588	Plicatura crispa	Plicatura crispa	1
Xylodon radula	9,612	Basidioradulum radula	Basidioradulum radula	1
Plagiochila asplenoides	8,956	Plagiochila asplenoides s.lat.	Plagiochila asplenoides s.str. Plagiochila asplenoides s.lat.	2
Stellaria apetala	8,274	Stellaria media	Stellaria media Stellaria pallida	2
Radula complanata	7,198	Radula complanata s.lat.	Radula complanata s.str. Radula complanata s.lat.	2
Lenzites betulinus	6,810	Trametes betulina	Trametes betulina	1
Senecio vernalis	6,644	Senecio leucanthemifolius subsp. vernalis	Senecio leucanthemifolius subsp. vernalis	1
Salix fragilis	6,424	Salix × fragilis	Salix × fragilis Salix euxina	2
Bubulcus ibis	5,296	Ardea ibis/coromanda	Ardea ibis/coromanda Ardea ibis	2

2. Tier 4 Matches (GBIF API)

These were resolved by querying the GBIF Species API for synonyms. The matches are generally reliable (the API provides the authoritative synonym chain), but it's good to scan the high-occurrence ones.

Top 50 by Occurrences

Table 4: 1,679 total Tier 4 matches — showing top 50

species	dyntaxa_scientificName	match_name_used	total_occ
Phylloscopus sibilatrix	Phylloscopus sibilatrix	motacilla sibilatrix	374,566
Acanthis hornemanni	Acanthis flammea	acanthis flammea	70,580
Thysselinum palustre	Peucedanum palustre	peucedanum palustre	42,506
Plebejus amanda	Polyommatus amandus	polyommatus amandus	39,016
Lolium pratense	Schedonorus pratensis	schedonorus pratensis	28,904
Parmeliella thriptophylla	Parmeliella triptophylla	parmeliella triptophylla	28,620
Lycæides idas	Plebejus argyrognomon	plebejus argyrognomon	26,904
Nyea lurideola	Eilema lurideola	eilema lurideola	23,356
Rabelera holostea	Stellaria holostea	stellaria holostea	21,993
Xanthoporia radiata	Mensularia radiata	mensularia radiata	21,574
Rubus polonicus	Rubus nessensis	rubus nessensis	21,069
Paracorymbia maculicornis	Stictoleptura maculicornis	stictoleptura maculicornis	18,717

species	dyntaxa_scientificName	match_name_used	total_occ
Symphytum uplandicum	Symphytum asperum	symphytum asperum	18,292
Calonarius odorifer	Cortinarius odorifer	cortinarius odorifer	17,589
Manulea complana	Eilema complana	eilema complana	17,486
Vertebrata byssoides	Vertebrata byssoides	brongniartella byssoides	17,390
Palaeochrysophanus hippothoe	Lycaena hippothoe	lycaena hippothoe	16,382
Hylotelephium maximum	Hylotelephium telephium	sedum telephium	16,046
Crambus nemorella	Crambus lathoniellus	crambus lathoniellus	14,078
Rhodocollybia asema	Rhodocollybia butyracea	rhodocollybia butyracea	13,620
Indalia lutarella	Eilema lutarella	eilema lutarella	13,396
Collybiopsis peronata	Gymnopus peronatus	gymnopus peronatus	13,266
Pararge petropolitana	Lasiommata petropolitana	lasiommata petropolitana	12,630
Amelanchier humilis	Amelanchier alnifolia	amelanchier alnifolia	12,528
Juncus gerardi	Juncus compressus	juncus compressus	12,170
Katha depressa	Eilema depressa	eilema depressa	11,846
Saxifraga osloensis	Saxifraga tridactylites	saxifraga tridactylites	11,624
Lolium giganteum	Schedonorus giganteus	schedonorus giganteus	11,490
Atheniella flavoalba	Mycena flavoalba	mycena flavoalba	11,439
Cratyna brevispina	Bradysia brevispina	bradysia brevispina	11,202
Cyanosporus caesius	Postia caesia	postia caesia	10,972
Spiraea billardii	Spiraea douglasii	spiraea douglasii	10,740
Podofomes mollis	Datronia mollis	datronia mollis	10,670
Mentha verticillata	Mentha aquatica	mentha aquatica	9,718
Protostropharia semiglobata	Stropharia semiglobata	stropharia semiglobata	9,573
Silene coronaria	Lychnis coronaria	lychnis coronaria	9,453
Silpha atrata	Phosphuga atrata	phosphuga atrata	9,387
Melanostoma mellina	Melanostoma certum	melanostoma dubium	9,308
Rhizosolenia delicatula	Guinardia delicatula	guinardia delicatula	9,260
Cerataulina bergonii	Cerataulina pelagica	cerataulina pelagica	9,194
Struthiopteris spicant	Blechnum spicant	blechnum spicant	8,750
Asterionella glacialis	Asterionellopsis glacialis	asterionellopsis glacialis	8,576
Mocyta fungi	Atheta fungi	atheta fungi	8,166
Plantago uliginosa	Plantago major	plantago major	8,048
Cotoneaster acutifolius	Cotoneaster lucidus	cotoneaster lucidus	7,671
Hieracium murorum	Hieracium scandinavium	hieracium silvaticum	7,646
Spiraea rosalba	Spiraea alba	spiraea alba	7,584
Bucklandiella heterosticha	Racomitrium canescens	racomitrium canescens	7,536
Orthodicranum flagellare	Dicranum flagellare	dicranum flagellare	7,430
Collybiopsis confluens	Gymnopus confluens	gymnopus confluens	7,377

Match Patterns

Pattern	Species	Occurrences
Different epithet (true synonym)	947	1,185,768
Genus reclassification	729	695,422
NA	3	19,396

3. Unmatched Species Analysis

Overview

Category	Species	Total Occurrences	Max Occurrences
binomial_no_match	3,799	593,270	108,618
genus_only	693	57,902	2,226
below_api_threshold	7,811	27,520	9
hybrid	6	548	460
other	2	184	178

Top Unmatched Binomials — Epithet Candidates

For the highest-occurrence unmatched binomials, we check if the **species epithet** exists in Dyntaxa under a different genus. These are the best candidates for manual resolution — likely genus reclassifications that the GBIF API missed or that fell below the API threshold.

```
## **5,712** unmatched species share an epithet with Dyntaxa:
## - **983** have exactly 1 Dyntaxa candidate (high confidence)
## - **4729** have multiple candidates (need manual review)
## - **5,898** have no epithet match in Dyntaxa at all
```

High-Confidence Candidates (1 Dyntaxa match)

These have a single Dyntaxa species sharing the epithet — almost certainly genus reclassifications. **These can likely be added as manual overrides directly.**

Table 7: 983 high-confidence epithet matches

gbif_species	dyntaxa_candidates	total_occ	match_category
Agriphila tristellus	Metriocnemus tristellus	20,212	binomial_no_match
Viola wittrockiana	Gloeochaete wittrockiana	6,380	binomial_no_match
Rhagio scolopacea	Limnodromus scolopaceus	5,684	binomial_no_match
Cortinarius mammillatus	Parasitus mammillatus	3,303	binomial_no_match
Lepisma saccharinum	Acer saccharinum	2,000	binomial_no_match
Lobesia fuligana	Pristerognatha fuligana	1,124	binomial_no_match
Anacaena globula	Euzetes globulus	849	binomial_no_match
Anasimyia interpunctus	Microterys interpunctus	792	binomial_no_match
Crocinitomix algicola	Collothea ambigua	664	binomial_no_match
Megaselia tenebricola	Tenuiphantes tenebricola	612	binomial_no_match
Spelobia quinata	Akebia quinata	582	binomial_no_match
Telenomus dalmani	Aegilips nitidula	578	binomial_no_match
Cortinarius exitiosus	Alternaria brassicae	543	binomial_no_match
Prorocentrum mexicanum	Sedum mexicanum	520	binomial_no_match
Rhamphomyia erythrothalma	Chromadorina erythrothalma	464	binomial_no_match
Pelagostrobilidium neptuni	Crossaster papposus	448	binomial_no_match
Hieracium glaucinum	Taraxacum glaucinum	400	binomial_no_match
Cotesia xyliana	Protothelenella xyliana	330	binomial_no_match

gbif_species	dyntaxa_candidates	total_occ	match_category
Pseudotontonia simplicidens	Spilichneumon simplicidens	284	binomial_no_match
Salix hookeriana	Deutzia hookeriana	274	binomial_no_match
Mydaea furtiva	Prosciara furtiva	272	binomial_no_match
Zoophthorus bridgmani	Promethes bridgmani	272	binomial_no_match
Plagiogramma brockmanni	Adlafia brockmannii	254	binomial_no_match
Dasyhelea malleolus	Teredora malleolus	250	binomial_no_match
Pleophragmia leporum	Pseudombrophila leporum	240	binomial_no_match
Scilla allenii	Erythrodermis traillii	234	binomial_no_match
Idris howardi	Anachipteria howardi	190	binomial_no_match
Chymomyza fuscimanus	Orthocladus fuscimanus	180	binomial_no_match
Inocybe mystica	Phaonia mystica	177	binomial_no_match
Tokunagaia kibunensis	Culicoides kibunensis	162	binomial_no_match
Fuchsia standishii	Thuja standishii	148	binomial_no_match
Maricaulis maris	Bicosoeca maris	146	binomial_no_match
Cricotopus osellai	Pholcus opilionoides	140	binomial_no_match
Gymnopus junquilleus	Neoboletus luridiformis var. junquilleus nom.prov.	135	binomial_no_match
Chalarus immanis	Vulgarogamasus immanis	134	binomial_no_match
Syntretus falcifer	Onchodellus falcifer	132	binomial_no_match
Chironomus nipponensis	Meteorus pulchricornis	130	binomial_no_match
Sourniaea diacantha	Callinectes sapidus	126	binomial_no_match
Paraphoma radicina	Alternaria radicina	126	binomial_no_match
Pseudoalteromonas ulvae	Peringia ulvae	118	binomial_no_match
Penicillium melinii	Cadophora fastigiata	116	binomial_no_match
Panaeolus linnaeanus	Gonioctena linnaeana	114	binomial_no_match
Aquirickettsiella gammari	Gammarinema gammari	114	binomial_no_match
Ctenochira oreophila	Euura freyja	114	binomial_no_match
Cortinarius tenebricus	Dinotoides tenebricus	105	binomial_no_match
Polycyclovorans algicola	Collothea ambigua	104	binomial_no_match
Tuberculatus remaudierei	Dendrocercus remaudierei	100	binomial_no_match
Chlamydodon triquetrus	Hylocomiadelpus triquetrus	96	binomial_no_match
Cratyna cryptospina	Platypalpus cryptospina	96	binomial_no_match
Bradysia pilistriata	Limnobaris dolorosa	96	binomial_no_match

Ambiguous Candidates (> 1 Dyntaxa match)

These need manual review — the epithet appears under multiple genera in Dyntaxa, so automated resolution is risky.

Table 8: 4729 ambiguous epithet matches

gbif_species	dyntaxa_candidates	n_candidates
Spilichneumon an-not-inum	Lycopodium annotinum Pohlia annotina	2 108,618

gbif_species_candidates	n_candidates
Trombidium maculata Anaspis maculata Rutpela maculata Siphona maculata Portevinia maculata Canthidermis maculata Miscogaster maculata Aleochara maculata Euophrys frontalis Dactylorhiza maculata Hypochaeris maculata Persicaria maculosa Nemophila lata maculata Euphorbia maculata Medicago arabica Potentilla crantzii Notonecta maculata Phyllodoce maculata Goniada maculata Myrianida pinnigera Lepidasthenia maculata Ophryotrocha maculata Pista maculata Gammaropsis maculata Ceropales maculata Stylochoplana maculata Rhodomonas maculata Rhodella violacea Dorylomorpha maculata Leucophenga maculata Spiniphora maculata Notiphila maculata Eucheilota maculata Graphomya maculata Paykullia maculata Psathyrella maculata Pterocallis maculata Macrocera maculata Idioptera maculata Ilisia maculata Rhipidia maculata Odinia trinitata Mycomya maculata Boletina maculata Phronia maculata Lyteba bisulca Ptychoptera contaminata Dixa maculata Eucoila maculata Lissonota maculata Diplocarpon mespili Pegomya maculata Rhodocollybia maculata Inocybe maculata/lanatodisca Mycena maculata Russula maculata Doto maculata Limnesia maculata Bryocrypta dubia Bathythrix maculata Inosperma maculatum Pomacea maculata Mestocharis maculata Lecanora impudens	64 36,020
Hieracium the lachenalii Carex lachenalii	2 14,948
Chaetochytrium fecunda Dinobryon sociale Tetrabaena socialis Oedogonium sociale	4 5,114
Podium humilis Tipula humilis Bombus humilis Stenus humilis Cyclodinus humilis Linepithema humile Aphelinus humilis Aprostocetus humilis Elachista humilis Araeonus humilis Lathys humilis Gnaphosa orites Erigeron humilis Scorzonera humilis Limonium humile Plagiobothrys calandrinoides Poa pratensis subsp. irrigata Baeomyces rufus Cladonia humilis Acrochaetium humile Crossopalpus humilis Suillia humilis Hydractinia humilis Gymnodia humilis Coenosia humilis Hyalopteroides humilis Semicytherura undata Potamocypris humilis Orchomene humilis Epidorylaimus humilis Phthiria humilis Blacus humilis Chelonus humilis Dicranella humilis Aulacoseira humilis Cuspidella humilis Inocybe humilis Tulipa humilis Dusona humilis Betula humilis Asterocarpa humilis	41 3,836
Chrysothrix bicinctus Paraplatypeza bicincta Exechia bicincta Anaretella defecta Lissonota bicincta Lecanora bicincta	6 3,506
Solidago terricola Margarinotus terricola Tropiphorus terricola Trochosa terricola Archaeodictyna consecuta Petaloproctus terricolus Protoblastenia terricola Chlorella vulgaris Tachydromia terricola Neurolyga excavata Eurystomina terricola Achromadora terricola Rhabditis aspera Pseudothielavia terricola Surirella terricola Llimoniella terricola Prodesmodora terricola Isogynandromyia terricola	18 3,028
Cinella pruinosa Schistidium pruinosa Xylodon detriticus Rhizocybe pruinosa Xenasma pruinosa	5 2,696
Cyrtus cornutum Hydroptila cornuta Tetramesa cornuta Hypomma cornutum Viola cornuta Proceraea cornuta Syllis cornuta Intoshella cornuta Polydora cornuta Cladonia cornuta Diastylis cornuta Eurycope cornuta Megamphopus cornutus Crisidia cornuta Eurylepta cornuta Spiniferomonas cornuta Eucampia cornuta Dohrniphora cornuta Stemonocera cornuta Turbanella cornuta Elachiptera cornuta Peromyia cornuta Acanthochondria cornuta Laophonte cornuta Semicytherura cornuta Tripyla cornuta Habrotrocha spicula Exechia cornuta Phronia cornuta Aceria cornuta Lecane cornuta Euura cornuta Boletina cornuta Arachnopeziza cornuta Usnea cornuta Filinia cornuta Osmia cornuta Astrorhizoides cornutus Ceratopteris cornuta	39 2,644

gbif_synonyms_candidates	n_candidates	total_dates
Priori Echinops exaltatus Lycopodium exaltatus	2	2,568
ex- al- ta- tus		
Lestodermella juniperina Vulpicida juniperinus Capronia juniperina Phaeocercospora juniperina Mycena juniperina Calycellina juniperina Gremmeniella juniperina Hysterostegiella juniperina Mollisia juniperina Mycosphaerella juniperina Scleropleella juniperina Stomiopeltis juniperina Tympanis laricina Gymnosporangium clavariiforme Lachnellula juniperina Oligotrophus juniperinus Tremella karstenii	17	2,558
Oligotrophus juniperinus Margaritella juniperina Parasola auricoma Coniochaeta hansenii Ascobolus hansenii Debaryomyces hansenii	5	2,404
Antennaria cameroni Cinetus cameroni Lyteba bisulca Trichopria cameroni Dicaelotus cameroni Euura vittata	6	2,146
Lacina araneosa Sphaerellothecium araneosum Sphaerellothecium araneosum var. araneosum Podospora araneosa Entoloma araneosum Trechispora araneosa	5	2,028
ne- osa		
Spiraea cinerea Formica cinerea Oplosia cinerea Brachyopa cinerea Motacilla cinerea Calandrella brachydactyla Donacia cinerea Micarea cinerea Phellodon violascens Zaira cinerea Xenus cinereus Potentilla incana Acronicta cinerea Agrotis cinerea Lepidochitona cinerea Arctosa cinerea Herniaria hirsuta subsp. cinerea Salix cinerea agg. Salix cinerea Nemoura cinerea Aspicilia cinerea Gyalecta russula Opegrapha vulgata Sporastatia polyspora Juglans cinerea Pompilus cinereus Agathomyia cinerea Megaselia cinerea Notiphila cinerea Leucophora cinerea Hydrotaea cinerea Fuligo cinerea Bolitophila cinerea Psychoda cinerea Lestremia cinerea Seimatosporium lichenicola Nepa cinerea Urosalpinx cinerea Draba cinerea Erica cinerea Arcyria cinerea Chaenotheca cinerea Basidioidendron cinereum Cerrena unicolor Clavulina cinerea Coprinopsis cinerea Ophiocordyceps entomorrhiza Mollisia cinerea Peniophora cinerea Palpomyia lineata Botrytis cinerea Haliclona cinerea Phytomyza cinerea Cuphophyllus lacmus	54	1,956
Neurospora tenuis terricola Margarinotus terricola Tropiphorus terricola Trochosa terricola Archaeodictyna consecuta Petaloproctus terricolus Protoblastenia terricola Chlorella vulgaris Tachydromia terricola Neurolyga excavata Eurystomina terricola Achromadora colaterricola Rhabditis aspera Pseudothielavia terricola Surirella terricola Llimoniella terricola Prodesmodora terricola Isogynandromyia terricola	18	1,790
Eudactylus biatorina Alchemilla propinqua Glaenocoris propinqua Gyptis propinqua Harmothoe propinqua Andrena propinqua Megaselia propinqua Hydrellia tarsata Harpinia propinqua Desoria propinqua Enhydrosoma propinquum Sarsameira propinqua Metopa propinqua Ctenochira propinqua Cacopsylla propinqua Pleuroceras propinquum Levinseniella propinqua Oribatula glabra	18	1,748
Gymnethetis autumnalis Harpalus autumnalis Tipula autumnalis Tabanus autumnalis Gastrancistrus autumnalis Spiranthes spiralis Scorzoneroidea autumnalis Callitriche hermaphrodita Adonis annua Syzygiella autumnalis Scytosiphon lomentaria Microcoleus autumnalis Musca autumnalis Dicranomyia autumnalis Mycetophila autumnalis Trimorus autumnalis Andricus quercusramuli Campodorus autumnalis Neotrombicula autumnalis Massariosphaeria autumnalis Apethymus filiformis Fimbristylis autumnalis Galerina marginata Hilversidia autumnalis Triplebea autumnalis Ophion autumnalis Dixella autumnalis Coregonus autumnalis	28	1,692
Hieracium vulgare Gnaphalium sylvaticum Equisetum sylvaticum Geranium sylvaticum Melampyrum sylvaticum Brachypodium sylvaticum Galium sylvaticum Plagiothecium vaticum Trichiosoma sylvaticum Globisporangium sylvaticum Hordelymus europaeus	11	1,686

gbif_synonyms_candidates	n_candidates	n_dates
Forcipomyia alacris Tenuiphantes alacris Pardosa alacris Lauritrioza alacris	4	1,426
Phaenocarpa sisypchia Zodion cinereum Leptolyngbya notata Cyanothece notata	4	1,338
Micropeltis varicolor Diadromus varicolor	2	1,226
Gryllotalpa vesicularis Thalloidima sedifolium Conops vesicularis Macrotrachela vesicularis Gypsina vesicularis Tholosina vesicularis	6	1,150
Brachydelphoides lenta Pholiota lenta Betula lenta Ochromonas lenta	4	1,114
Rhizopogon carbonaria Faerberia carbonaria	2	984
Cryptoglyphus foetens Entedon aestivalis Calosota aestivalis Adonis aestivalis Siphonurus aestivalis Aphelocheirus aestivalis Vezdaea aestivalis Agaricus altipes Dixella aestivalis Dichrogaster aestivalis Boletus reticulatus Bovista aestivalis Peronospora aestivalis Balsamia aestivalis	14	952
Russulachusa byssina Leptogium byssinum Myriococcum praecox Piloderma byssinum	4	932
Peniaria quercetorum Hebeloma quercetorum Hygrophorus cossus Entoloma quercetorum	4	890
Randallia jeffreysianus Sunius propinquus Cadulus propinquus Endococcus propinquus Cryptophilus propinquus Bryophaenocladus propinquus Dolichopus trivialis Microdeutopus quassanomalus Sarsamphiascus propinquus Orthopsyllus linearis Hippomedon propinquus Eusirus propinquus Bathylaimus inermis Enoplolaimus propinquus Syringolaimus tenuicaudatus Molophilus propinquus Atlantopandalus propinquus Amauronematus propinquus Euura caeruleocarpus Aeolothrips propinquus Haplothrips propinquus Cortinarius quarciticus Oribatula glabra Tubulicrinis propinquus	24	852
Gymnoclita japonica Torilis japonica Artemisia japonica Kerria japonica Reynoutria japonica Chaenomeles japonica Cryptomeria japonica Spiraea japonica Astilbe japonica Dasysiphonia japonica Perophora japonica Fibrocapsa japonica Skimmia japonica Delomerista japonica Styphnolobium japonicum Eriobotrya japonica Nuphar japonica Lemna japonica Pseudosasa japonica Aucuba japonica Coturnix japonica Pieris japonica Grandidierella japonica Trichocera japonica Actaea japonica Alternaria japonica Lonicera japonica Lopholeucaspis japonica Charybdis japonica Notholca japonica Paeonia japonica Rhabdomastix japonica Hylomecon japonica Arachnopeziza japonica Carpinus japonica Schizoporella japonica Nardia japonica Blyxa japonica Zostera japonica Saccharina japonica Ophryotrocha japonica Eriocheir japonica Anguilla japonica Popillia japonica	44	852
Randallia amplexicaulis Rudbeckia amplexicaulis	2	840

gbif_species_candidates	n_candidates
Hieracium cylindraceum Thesium alpinum Anthoxanthum nipponicum Eryngium alpinum alpinum Cerastium alpinum Laburnum alpinum Lactuca alpina Epimedium alpinum Lycopodium alpinum Phleum alpinum Trichophorum alpinum Woodsia alpina Aconogonon alpinum Thalictrum alpinum Ribes alpinum Synalissina isidioides Lichenostigma alpinum Stereocaulon alpinum Imbriobryum alpinum Diphascon alpinum Astiphromma alpinum Lophodermium alpinum Sphaeropezia alpina Austromenopon alpinum Neidium alpinum Antennaria alpina Colchicum alpinum Platyhypnum alpinum Polytrichastrum alpinum Cortinarius alpinus Hebeloma alpinum Cystopteris alpina Physarum alpinum Puccinia alpina Gymnomitrion alpinum Geoglossum alpinum Crocicreas alpinum Eusphalerum alpinum Diderma alpinum	39 834
Morrellia zonata Nereis zonata Leucia nivea Pyratula zonata Ulothrix zonata Megaselia zonata Pegomya zonata Arctoconopa zonata Enterographa zonata Polyplepta zonata Campoletis zonata Plectiscidea zonata Sagedia zonata Naucoria zonata Cercospora zonata Volucella zonaria	16 816
Megacelis abditus Erichthonius brasiliensis Tryphon abditus	3 814
Ostreococcus conicus Silene conica Microbryum davallianum var. conicum Cyclosa conica con-Amphisphaeria umbrina Senotainia conica Dasynemella conica Daptonema conicum ica Diapria conica Bicosoeca conica Schizothecium conicum Saemundssonina conica Entoloma velenovskii Hygrocybe conica Lycogala conicum Morchella conica Verpa conica Tritaxis conica Trachelomonas conica	19 794
Teleborus laeviceps Barycnemis bellator	2 742
Ablabesmus aspera Columella aspera Bromopsis benekenii Patella ulyssiponensis Centaurea as- aspera Poa glauca Chara aspera Harmothoe aspera Melanohalea exasperata Cymbella peraspera Megaselia leucozona Rhabditis aspera Scapania aspera Diatrypella aspera Euura sordidiapex Trochiscia aspera Amanita franchetii Xylodon asperus Echinoderma asperum Anthracoidea aspera Trachyneis aspera Hydrangea aspera Urceolella aspera	23 710
Sybinella obscurella Scoloposcelis obscurella Aleochara obscurella Aleochara punctatella ob- Bucculatrix humiliella Lecanora phaeostigma Javesella obscurella Phytomyza obscurella scuraeoneura obscurella Discocerina obscurella Mydaea obscurella Trichopalpus obscurellus Trioxoscelis obscurella Triaspis obscurella Hyposoter neglectus Scythris obscurella Caloplaca obscurella	17 702
Ectothenia californicus Euura myosotidis Neoseiulus californicus	3 702
Morrellia bryne turficola Lecidea turficola Psilocybe turficola Gymnopilus turficola Culicoides tur- fascipennis Atrichopogon minutus Dasyhelea turficola	7 670
fi- cola	

gbif species_candidates	n_candidates
Coturniculus suecicus Xyletinus planicollis Cypha suecica Phalacrus championi Dryocoetes sue-autographus Eupelmus pini Collentis suecicus Rubus suecicus Abrothallus suecicus ci- Calypogeia suecica Chaetocladius suecicus Prasanthus suecicus Enoploides suecicus cus Neognathorhynchus suecicus Bradylaimus suecica Sulcarius suecicus Mesochorus fulgurans Megastylus suecicus Scambus sagax Keratococcus suecicus Phacus suecicus Arrenurus suecicus Gaurax suecicus Lysitermus tritoma Sellnickochthonius suecicus Pseudopygmephorus suecicus Leptogamasus suecicus Domorganus suecicus Diplopeltoides suecicus Typhlodromus suecicus Guepiniopsis suecica	31 666
Astergyneta affinis Spilomena beata Trichonta beata	3 664
Ceratophyes decipiens Euplectus decipiens Amischa decipiens Cantharis decipiens de- Carpophilus ligneus Typhaea haagi Sitona lineellus Lactarius decipiens Limnephilus cip- decipiens Holcaeus decipiens Pteromalus eudecipiens Decipiphantes decipiens Sorbus × i- decipiens Salix × fragilis Saxifraga rosacea Setaria verticillata var. ambigua Calogaya ens decipiens Psora decipiens Thelidium decipiens Taraxacum decipiens Lamprodila decipiens Ahnfeltia plicata Daldinia decipiens Empoasca decipiens Thalassiosira decipiens Chaetoceros decipiens Auricula decipiens Protoperidinium decipiens Meiosimyza decipiens Celleporina decipiens Conisternum decipiens Pseudoterranova decipiens Pseudosinella decipiens Halichaetonotus dubius Proales decipiens Bdella longicornis Cinetus decipiens Quasimodoana decipiens Charops cantator Bathythrix decipiens Synodites decipiens Lecane decipiens Acrospermum graminum var. decipiens Oedogonium decipiens Rhypophila decipiens Ichneumon caedator Platylabus neglectus Trichosphaerella decipiens Pristiphora decipiens Brachysira decipiens Pterigynandrum filiforme Palustriella decipiens Athelia decipiens Physarum decipiens Cortinarius decipiens Gymnopilus decipiens Inocybe decipiens Russula decipiens Anema decipiens Anema nummularium Tilletia sphaerococca Ramularia rumicis Ilyocypris decipiens Scutellinia decipiens Ramularia simplex Melanconiella hyperopta Geotrichum decipiens Elaphomyces decipiens Quadriceps decipiens Hemitrichia decipiens Grimmia decipiens	71 656
Anomala longirostris Scolelepis squamata Scolelepis cantabra Apiognomonina hystrix Merziella lon-longirostris Bosmina longirostris Palaemon longirostris Helius longirostris Grimmia giro-longirostris Gnoriste longirostris Suctobelbella longirostris Aceria longirostris Bdelodes longirostris Ischnaspis longirostris Lulworthia longirostris Rostroculodes longirostris	16 652
Bryodistichium inclinatum Ptychostomum inclinatum Plagiostoma inclinatum Protoperidinium in- inclinatum	4 646
Umbellula cladiella dimorpha Perone dimorpha Tetradesmus dimorphus Siphogenerina dimorpha di- mor- pha	4 638
Spongiobolus labiatus Cryptocephalus labiatus Caliadurgus fasciatellus Acrobeloides nanus labi-Coronocycclus labiatus Notommata copeus	6 606
Boleophthalmus palmata Amphitrite cirrata Neoamphitrite affinis Palmaria palmata Melita pal- palmata Spelobia palmata Munna palmata Gammaropsis palmata Diclidophora palmata mata-Riccardia palmata Sasa palmata Thelephora palmata	12 584

gbif_dyntaxa_candidates	n_candidates
Pieskulomya maculata Anaspis maculata Rutpela maculata Siphona maculata Portevinia maculata Canthidermis maculata Miscogaster maculata Aleochara maculata Euophrys u- frontalis Dactylorhiza maculata Hypochaeris maculata Persicaria maculosa Nemophila lata maculata Euphorbia maculata Medicago arabica Potentilla crantzii Notonecta maculata Phyllodoce maculata Goniada maculata Myrianida pinnigera Lepidasthenia maculata Ophryotrocha maculata Pista maculata Gammaropsis maculata Ceropales maculata Stylochoplana maculata Rhodomonas maculata Rhodella violacea Dorylomorpha maculata Leucophenga maculata Spiniphora maculata Notiphila maculata Eucheilota maculata Graphomya maculata Paykullia maculata Psathyrella maculata Pterocallis maculata Macrocera maculata Idioptera maculata Ilisia maculata Rhipidia maculata Odinia trinotata Mycomya maculata Boletina maculata Phronia maculata Lyteba bisulca Ptychoptera contaminata Dixa maculata Eucoila maculata Lissonota maculata Diplocarpon mespili Pegomya maculata Rhodocollybia maculata Inocybe maculata/lanatodisca Mycena maculata Russula maculata Doto maculata Limnesia maculata Bryocrypta dubia Bathythrix maculata Inosperma maculatum Pomacea maculata Mestocharis maculata Lecanora impudens Leucolima pacifica Apus pacificus Samythella elongata Dolichovespula pacifica Gavia pacifica paci Thalassiosira pacifica Goniomonas pacifica Spilogona pacifica Candelaria pacifica fica Tomopteris pacifica Nardia pacifica Botrytella pacifica Chama pacifica Trepidosteptes pacificus	64 550

No Epithet Match

These species have no shared epithet with any Dyntaxa name — they are likely genuinely absent from the national backbone (foreign vagrants, non-native species, or taxa Dyntaxa doesn't cover).

Table 9: 5,898 species with no epithet in Dyntaxa — top 100

species	total_occ	match_type
Hieracium levicaule	12,256	binomial_no_match
Rhizosolenia fragillissima	10,610	binomial_no_match
Stenomacrus affinitor	3,436	binomial_no_match
Gymnometriocnemus kamimegavirgus	3,390	binomial_no_match
Catoptria margaritellus	2,298	binomial_no_match
Narcissus cyclazetta	2,288	binomial_no_match
Polietes lardaria	1,934	binomial_no_match
IMCC9063 sp000195085	1,790	binomial_no_match
Ctenolepisma longicaudatum	1,556	binomial_no_match
Cortinarius rufo-olivaceus	1,536	binomial_no_match
Rhagio tringaria	1,398	binomial_no_match
Megaselia arcticae	1,276	binomial_no_match
UBA9145 sp001438145	1,268	binomial_no_match
MS024-2A sp002384715	1,094	binomial_no_match
Umbelopsis gibberispora	1,044	binomial_no_match
Saitozyma podzolica	1,038	binomial_no_match
RS62 sp002340675	1,008	binomial_no_match
Linnemannia longigemmata	1,002	binomial_no_match
UBA2463 sp018402635	994	binomial_no_match
Epistrophella euchromus	982	binomial_no_match
Trioxys sunnysidensis	982	binomial_no_match
Acropiesta rufiventis	958	binomial_no_match

species	total_occ	match_type
UBA4582 sp002389265	950	binomial_no_match
Chrysoperla zastrowi	924	binomial_no_match
Conioscinella triorbiculata	920	binomial_no_match
MB11C04 sp002477515	906	binomial_no_match
Mortierella pseudozygospora	876	binomial_no_match
Phaonia luteva	864	binomial_no_match
CAIUKX01 sp016178305	858	binomial_no_match
HIMB59 sp902524245	852	binomial_no_match
Cinara pineum	836	binomial_no_match
AG-337-I02 sp902511405	820	binomial_no_match
CAILUG01 sp903887045	806	binomial_no_match
BACL21 sp002335345	776	binomial_no_match
SYDM01 sp905620505	772	binomial_no_match
BACL14 sp018607785	770	binomial_no_match
Lorryia volgini	750	binomial_no_match
Mortierella basiparvispora	746	binomial_no_match
Suctobelbella subcornigera	736	binomial_no_match
SYDM01 sp008638125	724	binomial_no_match
Inosperma ismeneanum	723	binomial_no_match
MS024-2A sp002292265	722	binomial_no_match
Orthocladus subletteorum	716	binomial_no_match
Cratyna keilini	714	binomial_no_match
Cortinarius heterodepressus	711	binomial_no_match
UBA9145 sp018401885	702	binomial_no_match
Platytes cerusella	698	binomial_no_match
Epermenia chaerophyllellus	692	binomial_no_match
Thelephora alnii	680	binomial_no_match
UBA11663 sp002313415	678	binomial_no_match

4. Manual Override Template

Export for Review

This section exports a CSV you can edit to add manual overrides. Fill in the `accept` column with `TRUE` for matches you've verified, and `dyntaxa_override` with the correct Dyntaxa name if the auto-suggested candidate is wrong.

```
## **Exported 1219 candidates** to:
## `/Users/lenathol2503/Desktop/gbif_data_gaps/data_proc/taxonomy_manual_overrides_template.csv`
##
## Edit this file, set `accept = TRUE` for verified matches, then 09a can
## read it as a Tier 0 override on the next run.
```

Table 10: Top 50 override candidates (full list in CSV)

gbif_species	suggested_dyntaxa	total_occ	source	accept	dyntaxa_override
Corvus cornix	Corvus corone cornix	2,245,881	tier2_ambiguous	NA	Corvus corone cornix

gbif_species	suggested_dyntaxa	total_source	acceptdyntaxa_	note
Ochlodes venata	Ochlodes sylvanus	83,564	tier2_ambiguous	NA
Rhodofomes roseus	Fomitopsis rosea	71,536	tier2_ambiguous	NA
Galium album	Galium mollugo subsp. erectum	34,112	tier2_ambiguous	NA
Inonotus leporinus	Pelloporus leporinus	23,914	tier2_ambiguous	NA
Agriphila tristellus	Metriocnemus tristellus	20,212	epithet_singular	NA
Plicaturopsis crispa	Plicatura crispa	17,588	tier2_ambiguous	NA
Xylodon radula	Basidioradulum radula	9,612	tier2_ambiguous	NA
Plagiochila asplenioides	Plagiochila asplenioides s.lat.	8,956	tier2_ambiguous	NA
Stellaria apetala	Stellaria media	8,274	tier2_ambiguous	NA
Radula complanata	Radula complanata s.lat.	7,198	tier2_ambiguous	NA
Lenzites betulinus	Trametes betulina	6,810	tier2_ambiguous	NA
Senecio vernalis	Senecio leucanthemifolius subsp. vernalis	6,644	tier2_ambiguous	NA
Salix fragilis	Salix × fragilis	6,424	tier2_ambiguous	NA
Viola wittrockiana	Gloeochaete wittrockiana	6,380	epithet_singular	NA
Rhagio scolopacea	Limnodromus scolopaceus	5,684	epithet_singular	NA
Bubulcus ibis	Ardea ibis/coromanda	5,296	tier2_ambiguous	NA
Pachykytospora tuberculosa	Haploporus tuberculosus	4,844	tier2_ambiguous	NA
Mycobilimbia sphaeroides	Mycobilimbia pilularis	3,608	tier2_ambiguous	NA
Cortinarius mammillatus	Parasitus mammillatus	3,303	epithet_singular	NA
Exidia cartilaginea	Exidia candida var. cartilaginea	2,202	tier2_ambiguous	NA
Lepisma saccharinum	Acer saccharinum	2,000	epithet_singular	NA
Apterygida albipennis	Apterygida media	1,882	tier2_ambiguous	NA
Sphaerodorum gracilis	Sphaerodorum flavum	1,882	tier2_ambiguous	NA
Atheta arctica	Philhygra arctica	1,776	tier2_ambiguous	NA
Scrippsiella hangoei	Apocalathium malmogiense	1,732	tier2_ambiguous	NA

gbif_species	suggested_dyntaxa	total_source	acceptdyntaxa_note	note
Atheta hygrobia	Philhygra hygrobia	1,644	tier2_ambiguous	NA Philhygra hygrobia
Gymnodinium simplex	Protodinium simplex	1,620	tier2_ambiguous	NA Protodinium simplex
Hapalopilus croceus	Aurantiporus croceus	1,456	tier2_ambiguous	NA Aurantiporus croceus
Lagis koreni	Pectinaria koreni	1,306	tier2_ambiguous	NA Pectinaria koreni
Xestia speciosa	Xestia baltica	1,230	tier2_ambiguous	NA Xestia baltica Xestia arctica
Atheta debilis	Philhygra debilis	1,227	tier2_ambiguous	NA Philhygra debilis
Mycenastrum corium	Mycenastrum corium (Lam. & DC.) Desv.	1,155	tier2_ambiguous	NA Mycenastrum corium (Lam. & DC.) Desv.
Lobesia fuligana	Pristerognatha fuligana	1,124	epithet_singular	NA NA NA
Atheta elongatula	Philhygra elongatula	1,068	tier2_ambiguous	NA Philhygra elongatula
Hymenoscyphus calyculus	Hymenoscyphus virgultorum	946	tier2_ambiguous	NA Hymenoscyphus virgultorum Hymenoscyphus serotinus
Atheta malleus	Philhygra malleus	918	tier2_ambiguous	NA Philhygra malleus
Neobulgaria pura	Ombrophila pura	886	tier2_ambiguous	NA Ombrophila pura
Bryum pallens	Ptychostomum pallens	854	tier2_ambiguous	NA Ptychostomum pallens
Anacaena globula	Euzetes globulus	849	epithet_singular	NA NA NA
Atheta palustris	Philhygra palustris	843	tier2_ambiguous	NA Philhygra palustris
Pachyphiale fagicola	Gyalecta fagicola	824	tier2_ambiguous	NA Gyalecta fagicola
Rhizopogon roseolus	Rhizopogon roseolus group	800	tier2_ambiguous	NA Rhizopogon roseolus group
Anasimyia interpunctus	Microterys interpunctus	792	epithet_singular	NA NA NA
Onnia triquetra	Pelloporus triquetra	766	tier2_ambiguous	NA Pelloporus triquetra
Melampsora caprearum	Melampsora farinosa	756	tier2_ambiguous	NA Melampsora farinosa
Erysiphe cichoracearum	Golovinomyces cichoracearum s.lat.	750	tier2_ambiguous	NA Golovinomyces cichoracearum s.lat. Golovinomyces cichoracearum
Delphinium ajacis	Consolida ajacis	732	tier2_ambiguous	NA Consolida orientalis Consolida ajacis
Epichloe typhina	Epichloë typhina s. lat.	730	tier2_ambiguous	NA Epichloë typhina s. lat. Epichloë typhina
Atheta britteni	Philhygra britteni	714	tier2_ambiguous	NA Philhygra britteni

How to Use Overrides

Once you've reviewed the CSV:

1. Open `data_proc/taxonomy_manual_overrides_template.csv`
2. For each row, set `accept = TRUE` if the suggested match is correct
3. If the suggestion is wrong but you know the correct name, fill in `dyntaxa_override`
4. Save as `data_proc/taxonomy_manual_overrides.csv`
5. A future version of 09a can read this file as a “Tier 0” pre-match step

5. Summary Statistics

```
## ## Reconciliation Health Check
##
## - **Total GBIF species:** 67,668
## - **Matched (Tiers 1-4):** 55,357 (81.8%)
## - **Occurrence coverage:** 99.8%
## - **Ambiguous matches needing review:** 236 (2,595,089 occ)
## - **Unmatched with epithet candidate:** 5,712
## - **Truly unresolvable:** 5,898
##
## ### Potential improvement from manual overrides

## If all **983** high-confidence epithet matches are accepted,
## occurrence coverage would increase from **99.76%**
## to **99.78%** (+59,257 occurrences).
```